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SQL Server on Azure Virtual Machines (IaaS) — concept to architecture, setup, management, cost, security, and best practices.

🧭 1. What is "SQL Server on Azure Virtual Machines (IaaS)"?

SQL Server on Azure VMs means running a **full SQL Server instance inside a Windows or Linux Virtual Machine** hosted in **Microsoft Azure's Infrastructure as a Service (IaaS)**.

It gives you:

- Full control over the OS and SQL Server.
- The same capabilities as on-prem SQL Server.
- Azure's flexibility, scalability, and pay-as-you-go pricing.

👉 **In short:**

It's like having your own on-prem SQL Server machine, but hosted in the Azure cloud.

🧩 2. The Deployment Model Context

Azure supports **three main models for running SQL workloads**:

Model	Description	Management Type
Azure SQL Database (PaaS)	Fully managed database only (no access to OS)	Microsoft manages everything
Azure SQL Managed Instance (PaaS)	Managed SQL instance with near 100% compatibility	Microsoft manages DB, user manages instance
SQL Server on Azure VM (IaaS)	Full SQL Server + OS control (your responsibility)	User manages OS + SQL

So, **IaaS** is chosen when you:

- Need full **SQL Server customization** (CLR, Agent Jobs, linked servers, etc.)
- Have **legacy applications** that need native SQL Server features.
- Want **direct OS access** (e.g., install antivirus, agents, 3rd party tools).
- Need **control over backups, HA/DR, and patching schedules.**

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🏗️ 3. Architecture Overview

Components:

1. **Azure Virtual Machine**
   - Runs Windows Server or Linux.
   - Configured with CPU, memory, and storage as per workload needs.
2. **SQL Server**
   - Installed manually or through Azure Marketplace images.
   - Versions: SQL Server 2016 → 2022 supported.
3. **Storage**
   - Azure Managed Disks (Premium SSD, Ultra Disk, or Standard SSD).
   - Data, logs, and TempDB ideally placed on **separate disks**.
4. **Networking**
   - Deployed inside a **Virtual Network (VNet)**.
   - Uses **Network Security Groups (NSGs)** for access control.
   - Optionally connected to on-prem via VPN or ExpressRoute.
5. **Management**
   - Azure **Automation**, **Backup**, and **Monitor** can integrate.
   - SQL IaaS Agent Extension simplifies management.

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⚙️ 4. Deployment Methods

You can deploy SQL Server VM in two main ways:

Option 1: Azure Marketplace Image

- Pre-installed with SQL Server.
- Choose edition: Developer, Web, Standard, or Enterprise.
- Automatically registers with SQL IaaS Agent Extension.

Option 2: Custom Installation

- Deploy a plain VM.
- Manually install SQL Server.
- Register SQL Server with **SQL IaaS Extension** for management.

🧰 5. SQL IaaS Agent Extension

This is key for managing SQL inside Azure.
It provides Azure-level management features like:

Category	Capabilities
Licensing	Pay-as-you-go or Bring Your Own License (BYOL)
Automated Backup	Schedule and retention for full backups
Automated Patching	Control patch window for OS + SQL
Azure Monitoring	Performance metrics, alerts, insights
Security Configuration	Encryption, Azure Key Vault integration
HA/DR Support	Integration with Always On Availability Groups

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You can verify or enable it via the Azure Portal → SQL Virtual Machines → Configuration.

🧩 6. High Availability & Disaster Recovery (HA/DR)

Options available:

1. **Always On Availability Groups (AGs)**
   - High availability + read-scale.
   - Requires multiple VMs and Windows Failover Clustering.
2. **Failover Cluster Instance (FCI)**
   - Shared disk setup for SQL HA.
   - Use Azure Shared Disks.
3. **Log Shipping / Database Mirroring**
   - Simpler DR setups.
4. **Azure Site Recovery (ASR)**
   - Replicates the entire VM for disaster recovery.

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🔐 7. Security Considerations

Area	Recommendation
Network Access	Use private IPs and NSGs; avoid public ports.
Authentication	Use Azure AD authentication or Managed Identity.
Encryption	Enable Transparent Data Encryption (TDE).
Backups	Store in Azure Storage (encrypted with Key Vault keys).
Patching	Automate through Azure Update Management.

📊 8. Monitoring & Maintenance

Azure provides several integrated options:

- **Azure Monitor / Log Analytics** – collects performance and logs.
- **SQL Insights** – deep SQL-level telemetry.
- **Azure Automation** – for custom maintenance jobs (index rebuilds, etc.).
- **Azure Backup** – for VM-level or database-level protection.

💰 9. Costing Model

Two major cost factors:

Component	Description
VM Compute	CPU, RAM, and uptime.
Storage	Managed disks (data, log, backup).
SQL License	Pay-as-you-go or BYOL (Azure Hybrid Benefit).

Example: If you have an existing SQL Server Enterprise license with Software Assurance, you can **save ~50%** using **Azure Hybrid Benefit**.

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🚀 10. Performance Best Practices

1. Use **Premium SSD** for Data & Log files.
2. Separate **Data, Log, TempDB** onto different disks.
3. Enable **Read/Write caching** carefully (Data = Read-only, Log = None).
4. Use **Accelerated Networking** for better throughput.
5. Configure **MAXDOP, TempDB**, and memory limits as per workload.
6. Use **Storage Spaces** if you need striping for IO performance.

🧪 11. Real-World Example

Imagine you're migrating an **on-prem ERP database** (500 GB) to Azure.

Steps:

1. Create a **Windows Server 2022 VM** with 8 vCPUs, 32 GB RAM.
2. Deploy **SQL Server 2022 Enterprise** from Azure Marketplace.
3. Attach 3 Managed Disks:
   - D:\ for Data
   - E:\ for Logs
   - F:\ for TempDB
4. Join it to your **domain (Active Directory)**.
5. Configure **Always On AG** with a secondary VM.

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6. Enable **Automated Backup** and **Monitoring via SQL IaaS Extension**.
7. Use **Azure Bastion** for secure RDP access.

🧠 12. When to Choose SQL Server on Azure VM

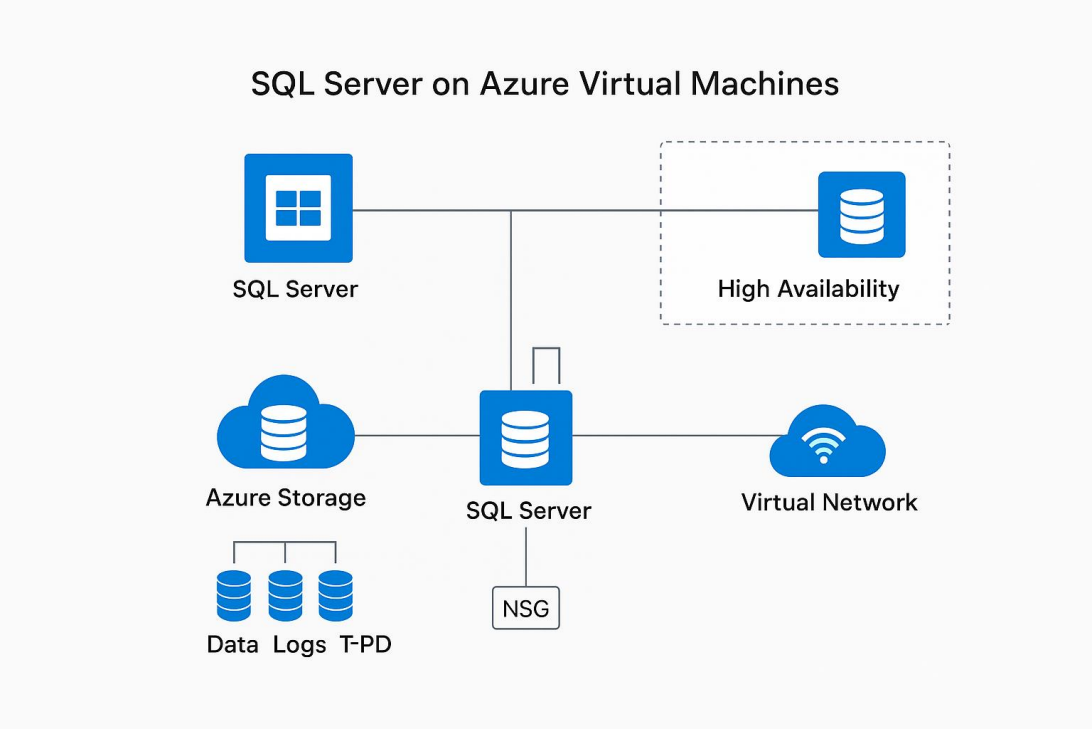
✅ Choose this if:

- You need full OS & SQL control.
- Your app requires features not available in PaaS (e.g., cross-instance queries, custom agents, FILESTREAM).
- You're migrating "as-is" from on-prem without redesign.

❌ Avoid if:

- You prefer Microsoft to handle patching, backup, HA automatically (choose **SQL Managed Instance**).
- You need simple, elastic scale (choose **Azure SQL Database**).

📘 Summary

Aspect	Description
Service Type	Infrastructure as a Service (IaaS)
Management Responsibility	Customer manages OS & SQL
Deployment	Azure VM + SQL Server
Licensing	PAYG or BYOL
Best For	Full SQL control, legacy apps, custom configurations
HA/DR	AGs, FCI, ASR supported
Monitoring	Azure Monitor + SQL IaaS Extension

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